Hypothesis Testing and Decision Errors

Lectures 20–21

QBA201 – Quantitative Business Analysis

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Set up the factory's question

The factory aims for 500 mL. Let μ be the population mean fill.

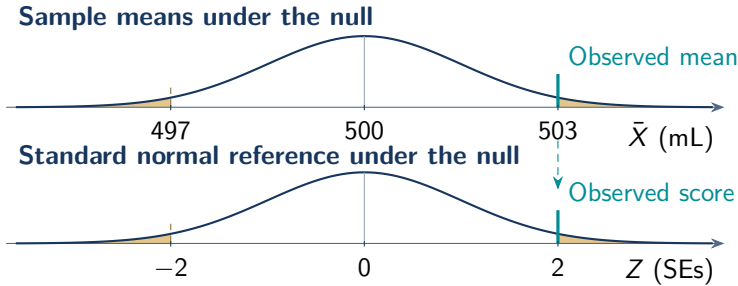
$$H_0:\ \mu = 500\ \text{mL} \qquad H_1:\ \mu \neq 500\ \text{mL}$$

Underfilling and overfilling both matter. Choose a two-sided test at $\alpha = 0.05$ before looking at the sample.

The observed mean is $\bar{x} = 503$ mL. How unusual is this gap if the process is on target?

One null model on two scales

Independent normal fills, $n = 64$, known $\sigma = 12$ mL. Under H_0, the mean of $\bar{X}$ is 500 mL and its standard error is $12/\sqrt{64} = 1.5$ mL.



$$z_{\text{obs}} = \frac{503 - 500}{1.5} = 2 \qquad \underbrace{\mathrm{P}_{H_0}(|\bar{X} - 500| \geq 3)}_{\text{mL scale}} = \underbrace{\mathrm{P}(|Z| \geq 2)}_{\text{standardized scale}} \approx 0.0455$$

Make the decision and report the evidence

The **p-value** is the probability, under H_0, of a result at least as extreme as the one observed in the direction of H_1.

$$\underbrace{0.0455}_{\text{observed p-value}} < \underbrace{0.05}_{\text{chosen significance level}} \implies \text{Reject } H_0.$$

Report evidence that the mean fill differs from 500 mL. The cost of adjusting the machine is a separate business decision.

If $p > \alpha$, fail to reject. The evidence is inconclusive.

Read the decision as a cutoff

Let z_q be the standard normal cutoff with area q above it. At 5% two-sided, set aside 2.5% in each tail.

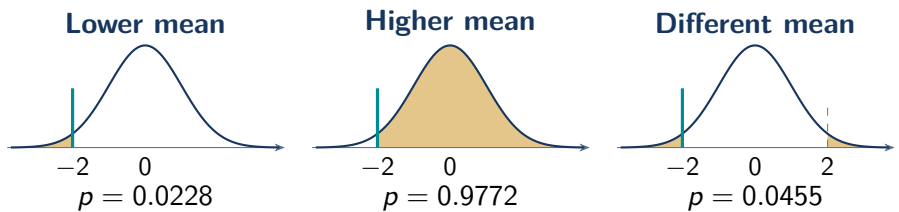
$$\text{Reject for } |z_{\text{obs}}| \geq z_{0.025} \approx 1.96.$$

Range	Bottling example
Sample means within the cutoffs	$500 \pm 1.96(1.5) = [497.06, 502.94]$
95% confidence interval for μ	$503 \pm 1.96(1.5) = [500.06, 505.94]$

The observed mean 503 is outside the first range. The target 500 is outside the second. Both lead to rejection.

The alternative tells you which area to count

Compare three *preplanned* questions at the same observed $z = -2$.



$\Phi(z_{\text{obs}})$ (left) $\quad 1 - \Phi(z_{\text{obs}})$ (right) $\quad 2[1 - \Phi(|z_{\text{obs}}|)]$ (two-sided)

Pick the tail from the business question before looking at the sample.

Write the claim, then use the matching cutoff

Let c be the benchmark mean. For a one-sided test, calculate the null reference distribution at $\mu = c$.

Business question	H_0	H_1	Reject when
Is the mean lower?	$\mu \geq c$	$\mu < c$	$z_{\text{obs}} \leq -z_\alpha$
Is the mean higher?	$\mu \leq c$	$\mu > c$	$z_{\text{obs}} \geq z_\alpha$
Is the mean different?	$\mu = c$	$\mu \neq c$	$\lvert z_{\text{obs}} \rvert \geq z_{\alpha/2}$

The magnitude 1.645 leaves 5% in one tail. It also marks each boundary of a two-sided test at $\alpha = 10\%$.

Apply the left-tail rule to a salary claim

Do comparable jobs pay *less than* \$180,000 on average? Choose $H_0 : \mu \geq 180{,}000$, $H_1 : \mu < 180{,}000$, and $\alpha = 0.05$.

An independent sample of 100 jobs has $\bar{x} = 174{,}600$, with known $\sigma = 30{,}000$.

$$\text{SE} = 3{,}000, \qquad z_{\text{obs}} = \frac{174{,}600 - 180{,}000}{3{,}000} = -1.8, \qquad p = 0.0359.$$

Reject H_0. The sample provides evidence of a mean below \$180,000 for the population sampled.

Average pay and the share of high earners

Two managers ask different questions about a salary benchmark of \$200,000.

Is average pay higher?

Use the population mean, μ.
A claim of higher average pay gives $H_1 : \mu > 200{,}000$.

Do few salaries exceed the benchmark?

Use the population proportion, p.
Count salaries above \$200,000 and specify what rate qualifies as few.

The business question determines which population feature to test.

The question determines the standard error

Let d_0 be the null difference. For every row, use $z = (\text{estimate} - \text{null value})/\text{null standard error}$.

Target	Estimate and null value	Standard error for the test
One mean, known σ	$\bar{x},\ \mu_0$	$\sigma/\sqrt{n}$
One proportion	$\hat{p},\ p_0$	$\sqrt{p_0(1-p_0)/n}$
Difference in means	$\bar{x}_1 - \bar{x}_2,\ d_0$	$\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$
Difference in proportions	$\hat{p}_1 - \hat{p}_2,\ 0$	$\sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})(1/n_1 + 1/n_2)}$

For the difference rows, use independent groups. Mean SDs are known. In the equality test of two rates, $\hat{p}_{\text{pool}}$ is total successes divided by total observations.

What can go wrong when we use the rule repeatedly?

Decision	H_0 true	H_0 false
Reject H_0	Type I: false alarm	Detect a departure
Fail to reject H_0	Correct nonrejection	Type II: miss a departure

The chance of a miss, β, depends on the true alternative mean.

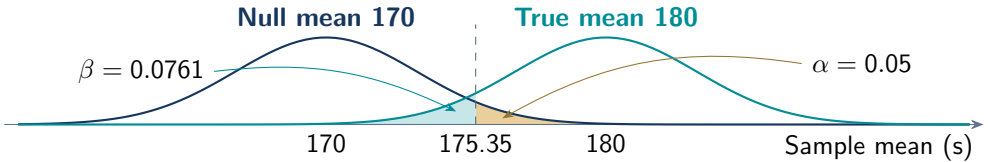
Evaluate a rule before the data arrive

If the factory prescribes an alarm at $|\bar{X} - 500| \geq 3$ mL, the null graph gives a false-alarm rate of 4.55% for that rule.

The same area served as the p-value for the observed 503 mL. The Type I rate evaluates a fixed rule over repeated samples.

One cutoff, two possible process means

Billing time example. Test whether the population mean exceeds 170 seconds. The standard error of the sample mean is 3.25 seconds.



Cutoff $= 170 + 1.6449(3.25) \approx 175.35$ Power at $180 = 1 - 0.0761 = 0.9239$

A stricter alarm trades off false alarms and detection

Keep the billing example's true mean at 180 seconds and its standard error at 3.25 seconds. Let β be the probability of missing this slowdown.

False-alarm rate α	Cutoff (s)	Miss rate β	Power $1-\beta$
5%	175.35	7.61%	92.39%
1%	177.56	22.65%	77.35%

At a fixed false-alarm rate, more independent observations can improve detection by reducing $\mathrm{SE} = \sigma/\sqrt{n}$.